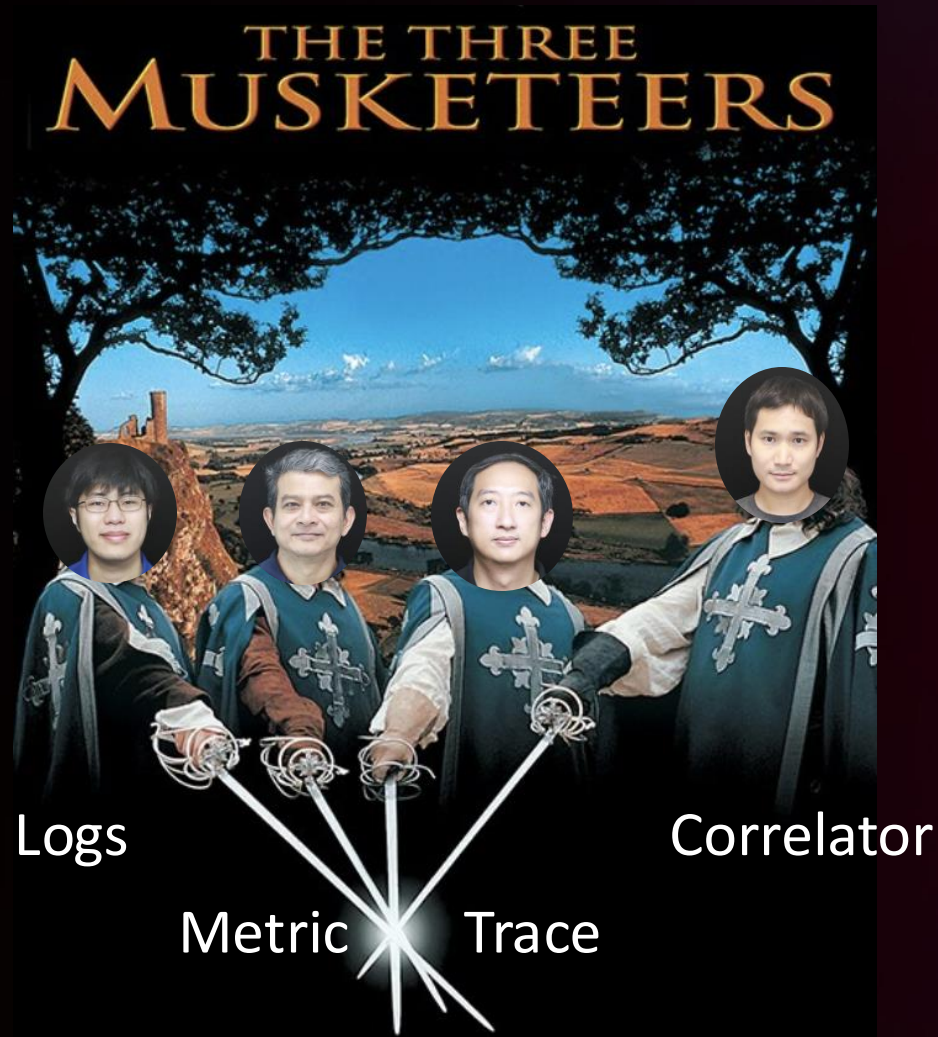


Observability-Driven Automation with OpenTelemetry and AI for Operation

Observability Instrumentation

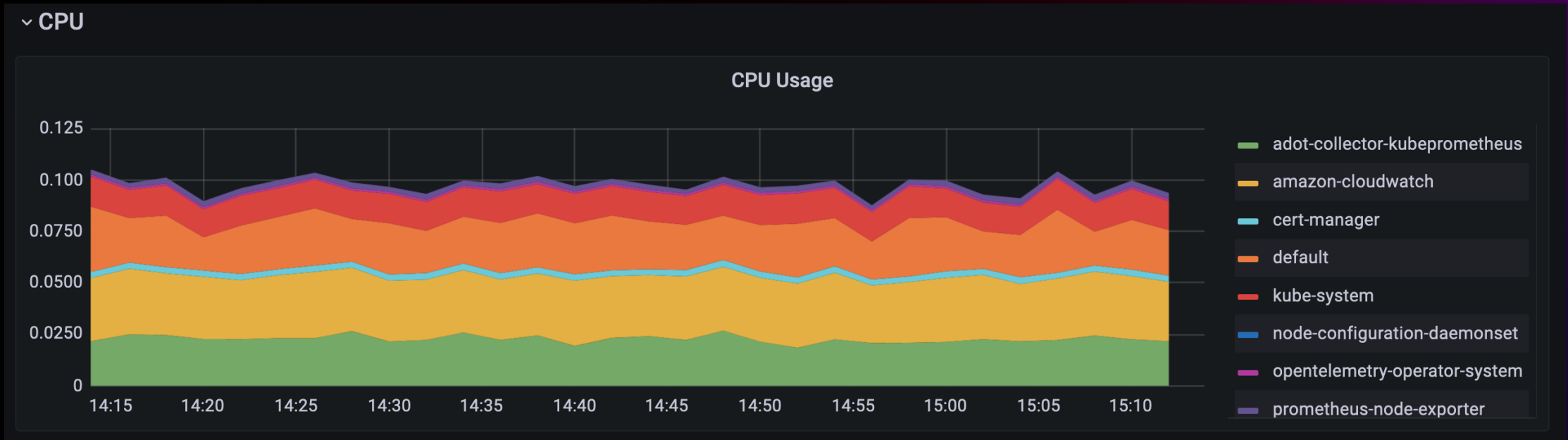


Agenda

- Observability Instrumentation
- Logging in Kubernetes
- Metrics in Kubernetes
- Traces in Kubernetes
- AI in Observability

Metrics

Time series measurement of system performance, health or business data over a period.
Typically visualized on graphical charts for human consumption



Logs

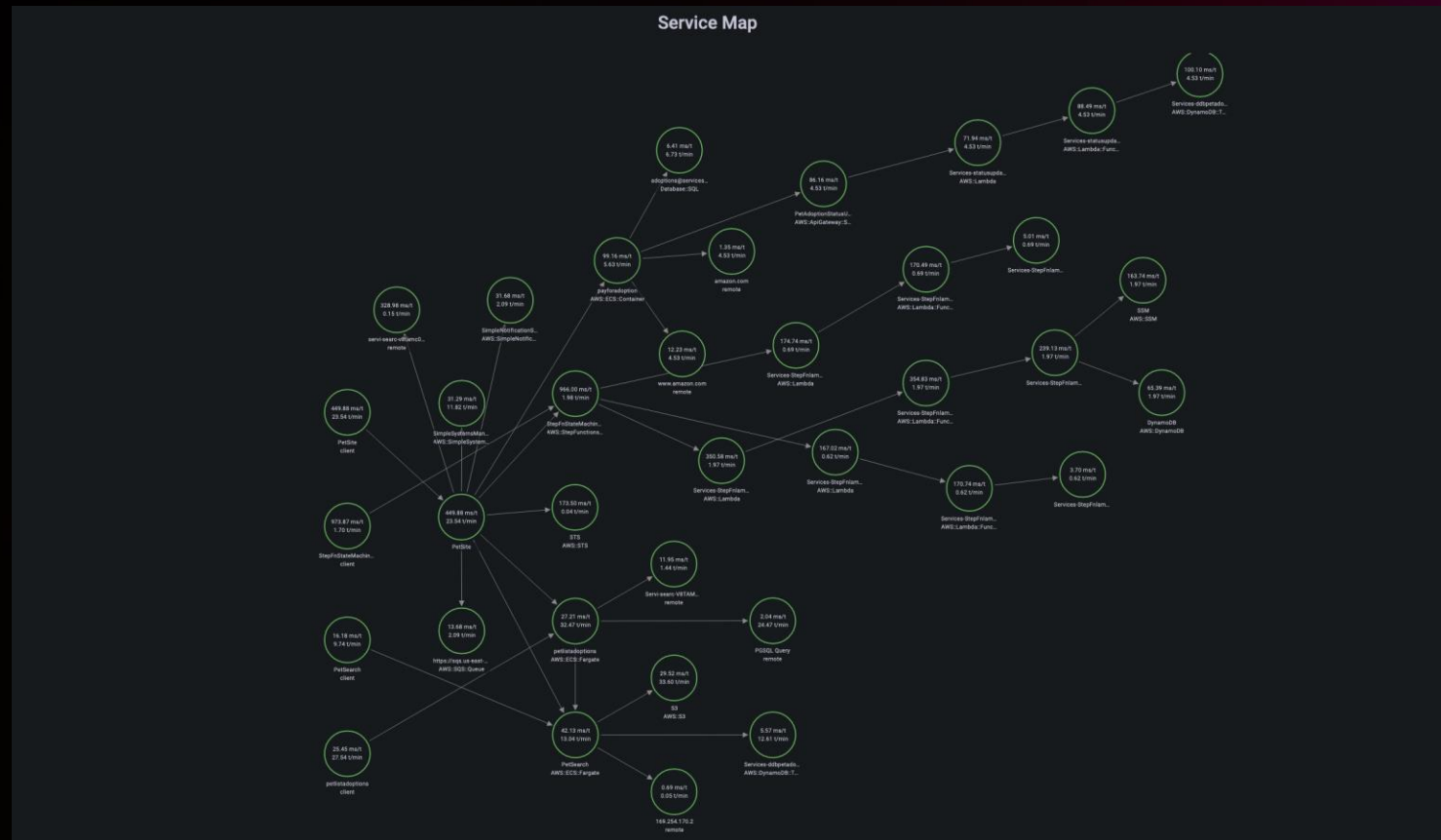
Textual data captured from applications indicating an event happened at that time.

Log events are queried to perform deep analysis or troubleshooting

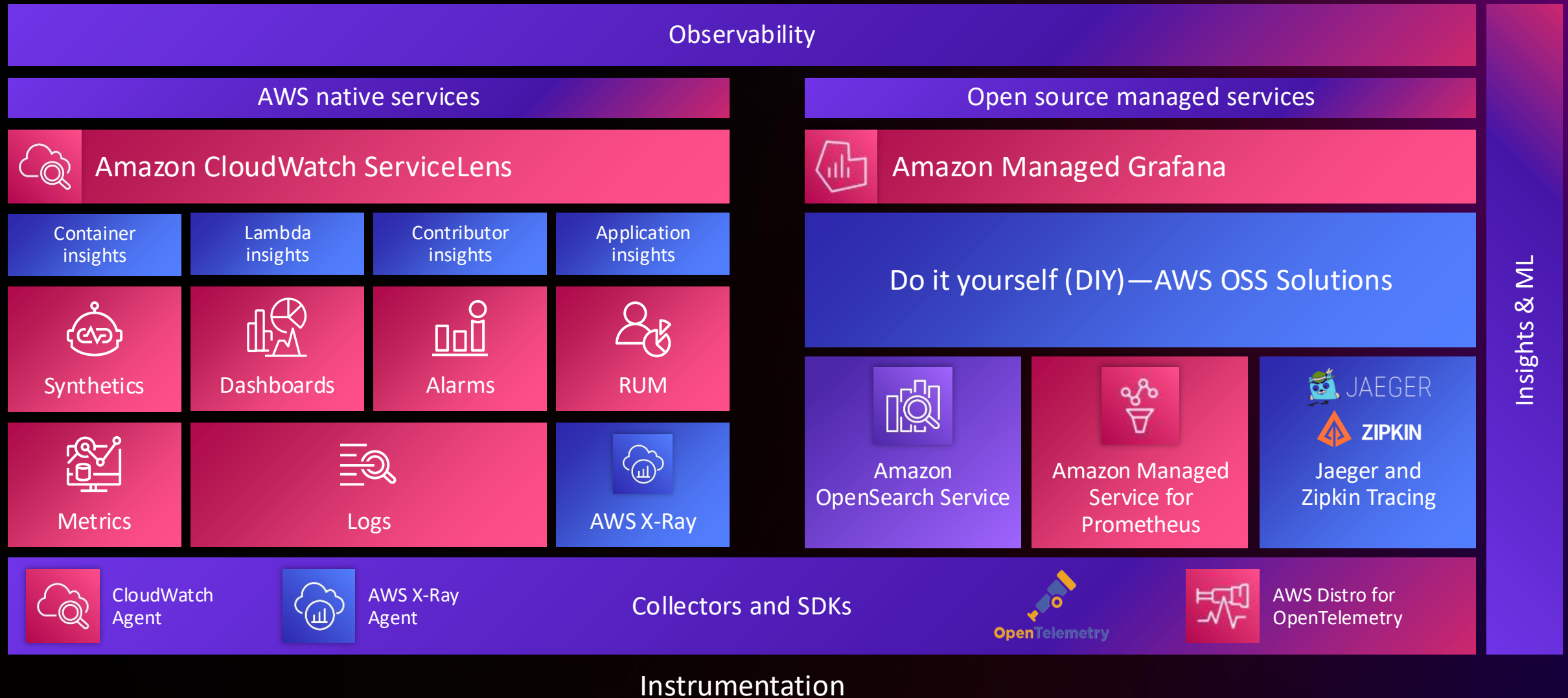
[eCommerce] Orders				
1-50 of 1034 < >				
Time ▾	category	sku	taxful_total_price	total_quantity
> Oct 14, 2022 @ 10:57:07.000	Women's Clothing	ZO0060600606, ZO0052600526	\$49.98	2
> Oct 14, 2022 @ 10:47:02.000	Women's Shoes	ZO0371303713, ZO0030600306	\$66.98	2
> Oct 14, 2022 @ 10:38:24.000	Men's Clothing	ZO0583405834, ZO0458504585	\$45.98	2
> Oct 14, 2022 @ 10:19:41.000	Women's Clothing	ZO0265702657, ZO0706107061	\$49.98	2
> Oct 14, 2022 @ 10:13:12.000	Men's Clothing	ZO0130201302	\$6.99	1
> Oct 14, 2022 @ 10:03:50.000	Women's Clothing, Women's Shoes	ZO0032600326, ZO0014000140	\$49.98	2
> Oct 14, 2022 @ 09:56:38.000	Women's Shoes, Women's Clothing	ZO0019900199, ZO0643606436	\$52.98	2
> Oct 14, 2022 @ 09:49:26.000	Men's Shoes, Men's Clothing	ZO0690206902, ZO0297302973, ZO0565905659, ZO0621806218	\$153.96	4
> Oct 14, 2022 @ 09:43:41.000	Women's Shoes	ZO0019300193, ZO0367503675	\$79.98	2
> Oct 14, 2022 @ 09:43:41.000	Men's Accessories, Men's Clothing	ZO0467104671, ZO0424404244, ZO0571305713, ZO0274602746	\$223.96	4
> Oct 14, 2022 @ 09:37:55.000	Women's Shoes	ZO0367603676, ZO0704407044	\$154.98	2

Traces

A record of the end-to-end journey of user requests, from the user-interface through the entire distributed system and back to the user.



AWS Observability options



Logging in K8s



K8s control plane logs in CloudWatch Logs

Kubernetes API server component logs (api) – Your cluster's API server is the control plane component that exposes the Kubernetes API

Audit (audit) – Kubernetes audit logs provide a record of the individual users, administrators, or system components that have affected your cluster

Authenticator (authenticator) – Authenticator logs are unique to K8s; these logs represent the control plane component that uses for Kubernetes role-based access control (RBAC) authentication using IAM credentials

Controller manager (controllerManager) – The controller manager manages the core control loops that are shipped with Kubernetes

Scheduler (scheduler) – The scheduler component manages when and where to run pods in your cluster

Logging and Auditing the Control Plane

Logging of the control plane, especially around an **audit trail** of API actions, is an important aspect of security and being able to work out who did what.

When using EKS you can (and should) enable such logging to CloudWatch Logs.

Logging			Update
CloudWatch /aws/eks/cluster/cluster	API server Enabled	Audit Enabled	
Authenticator Enabled	Controller manager Enabled	Scheduler Enabled	

Different ways to collect application logs in your cluster

A common pattern using fully open tooling is collecting logs with **fluentd/fluent-bit** to stream directly into an **Elasticsearch** where you'll query/visualize them with **Kibana** – this is often called the EFK stack.

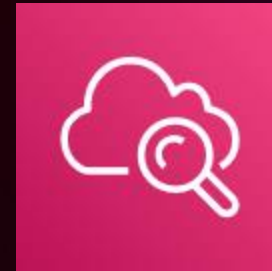
While you can run all these components yourself on your cluster, we'd encourage you to use the managed Amazon Elasticsearch (which also includes Kibana).

You can also use **fluentd/fluent-bit** to send the logs to **CloudWatch Logs**. This may entirely meet your needs or you can also optionally stream them elsewhere too.



Elasticsearch

/ Amazon Elasticsearch Service

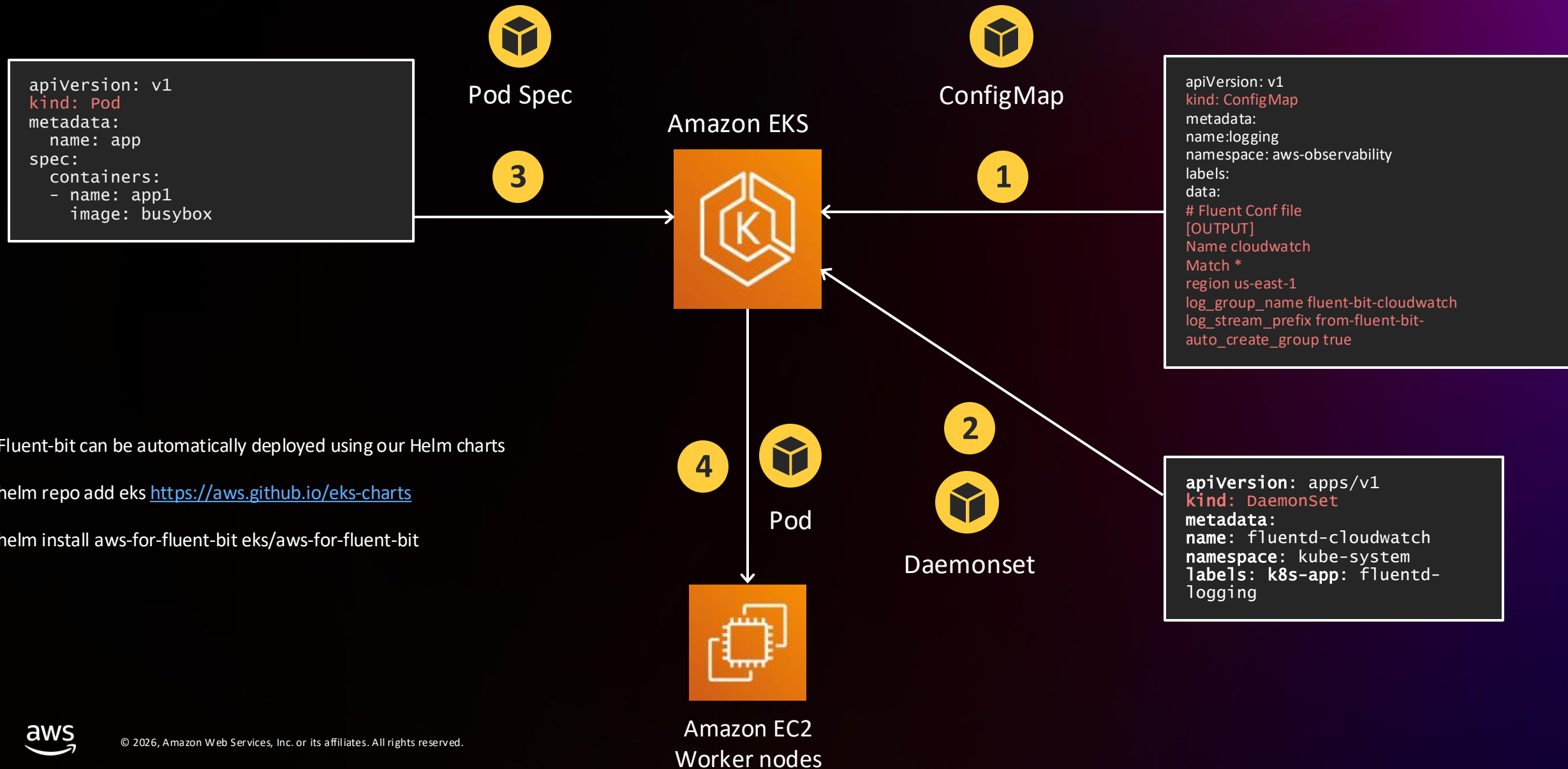


Amazon CloudWatch

Simplified using aws-for-fluent-bit Helm Chart

<https://aws.github.io/eks-charts>

Logging on EKS



Metrics in K8s



CHALLENGE:

Increasing management
and cost overhead

Increased time to resolve issues as
metrics

SOLUTION:

Standardized open source APIs,
SDKs, and agents

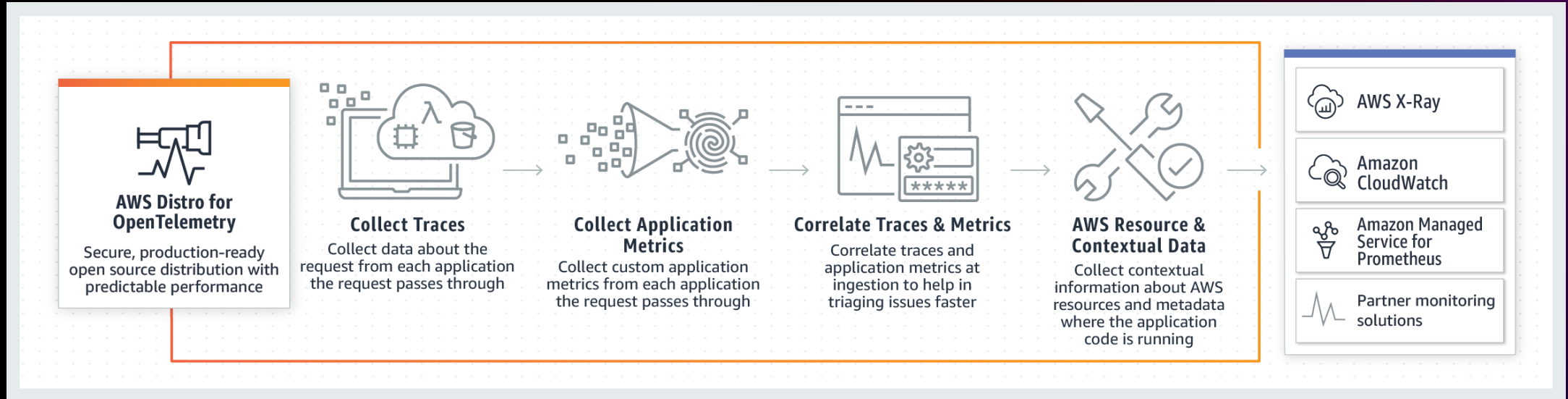
Correlate metrics and traces
at ingestion along with contextual
information, reducing time to
resolution

OpenTelemetry provides...

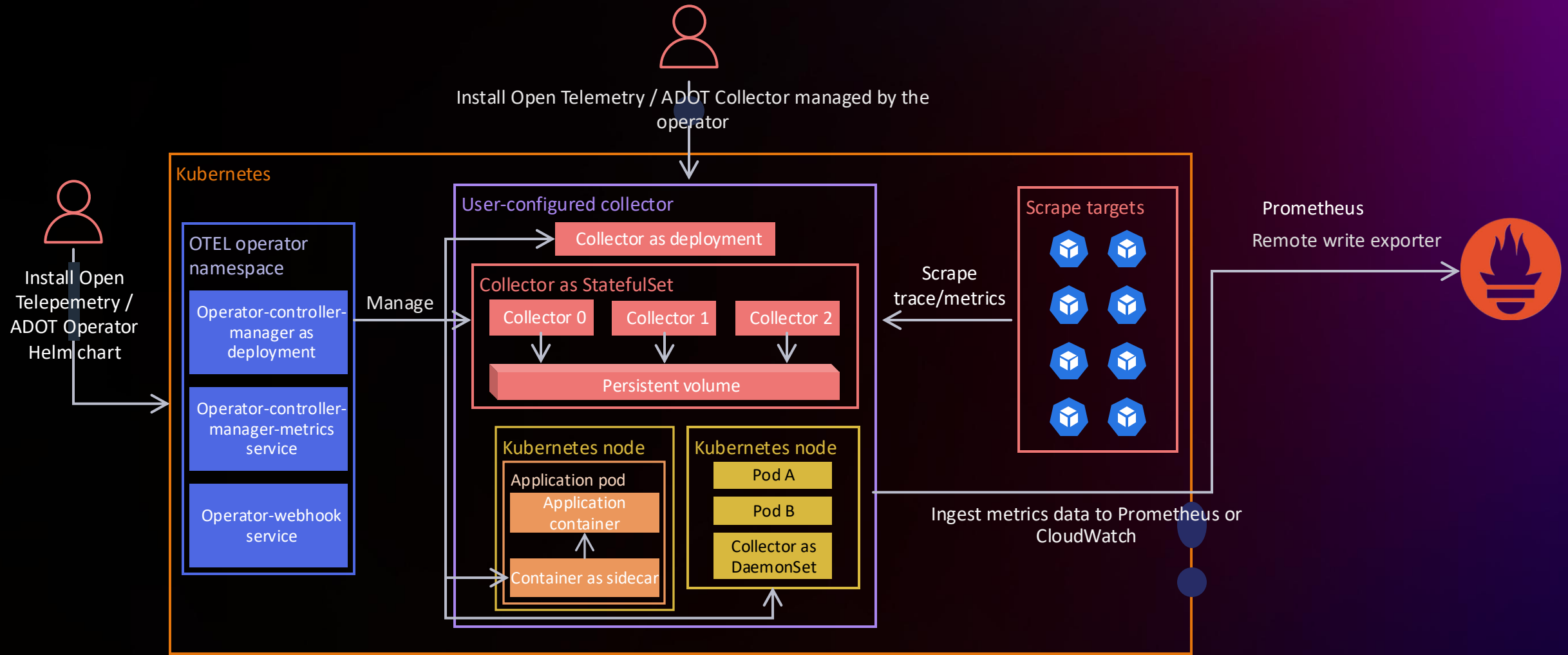
- A single, vendor-agnostic instrumentation library per language with support for both automatic and manual instrumentation
- A single collector binary that can be deployed in a variety of ways
- An end-to-end implementation to generate, emit, collect, process and export telemetry data.
- Full control of your data with the ability to send data to multiple destinations in parallel through configuration
- Open-standard semantic conventions to ensure vendor-agnostic data collection
- The ability to support multiple context propagation formats in parallel to assist with migrating as standards evolve
- A path forward no matter where you are on your observability journey. With support for a variety of open-source and commercial protocols, format and context propagation mechanisms as well as providing shims to the OpenTracing and OpenCensus projects, it is easy to adopt OpenTelemetry.

AWS Distro for OpenTelemetry

AWS Distro for OpenTelemetry is a secure, production-ready, AWS-supported distribution of the OpenTelemetry project.



OpenTelemetry Kubernetes operator



Traces in K8s



Use Cases



Analyze and debug applications

Receive trace data from your simple and complex applications, whether they are in development or production.



Generate a detailed service map

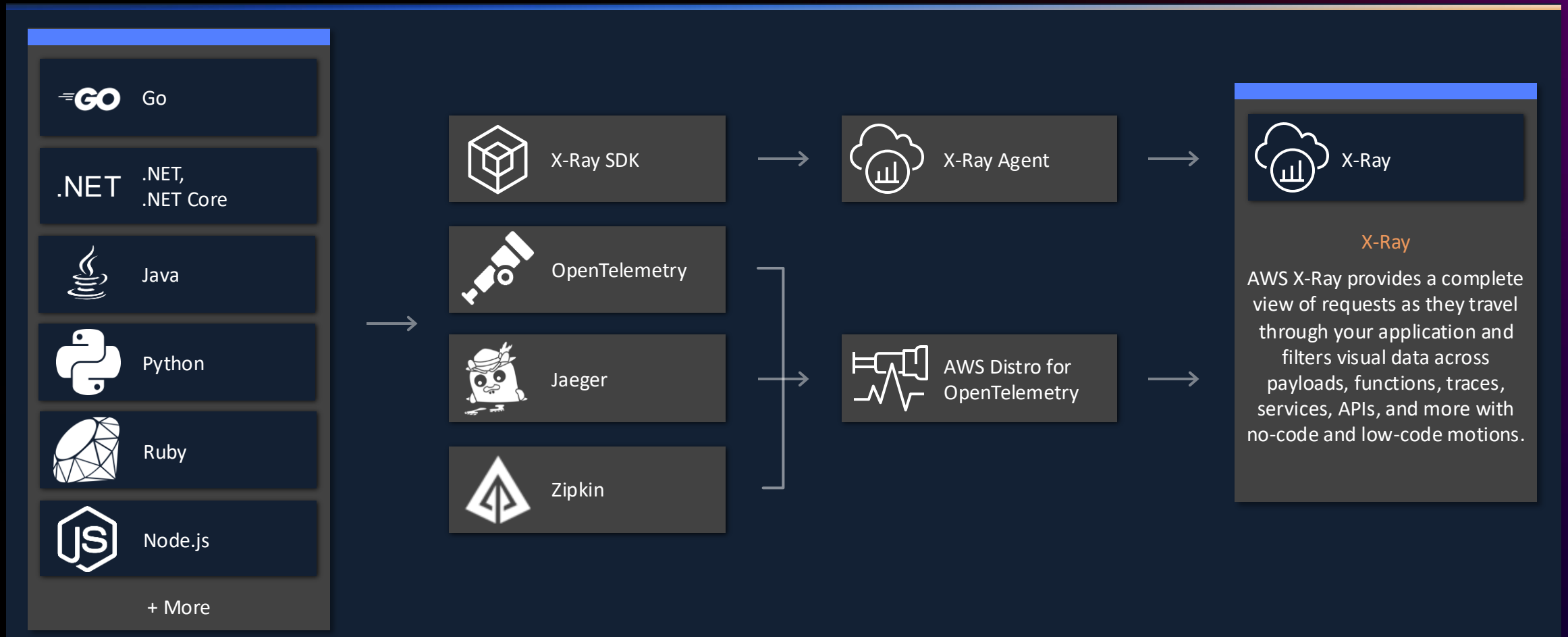
Compile data from your AWS resources to determine bottlenecks in your cloud architecture and improve application performance.



View performance analytics

Compare trace sets with different conditions for root cause analysis purposes.

Choice



Open Telemetry Endpoint (OTLP)

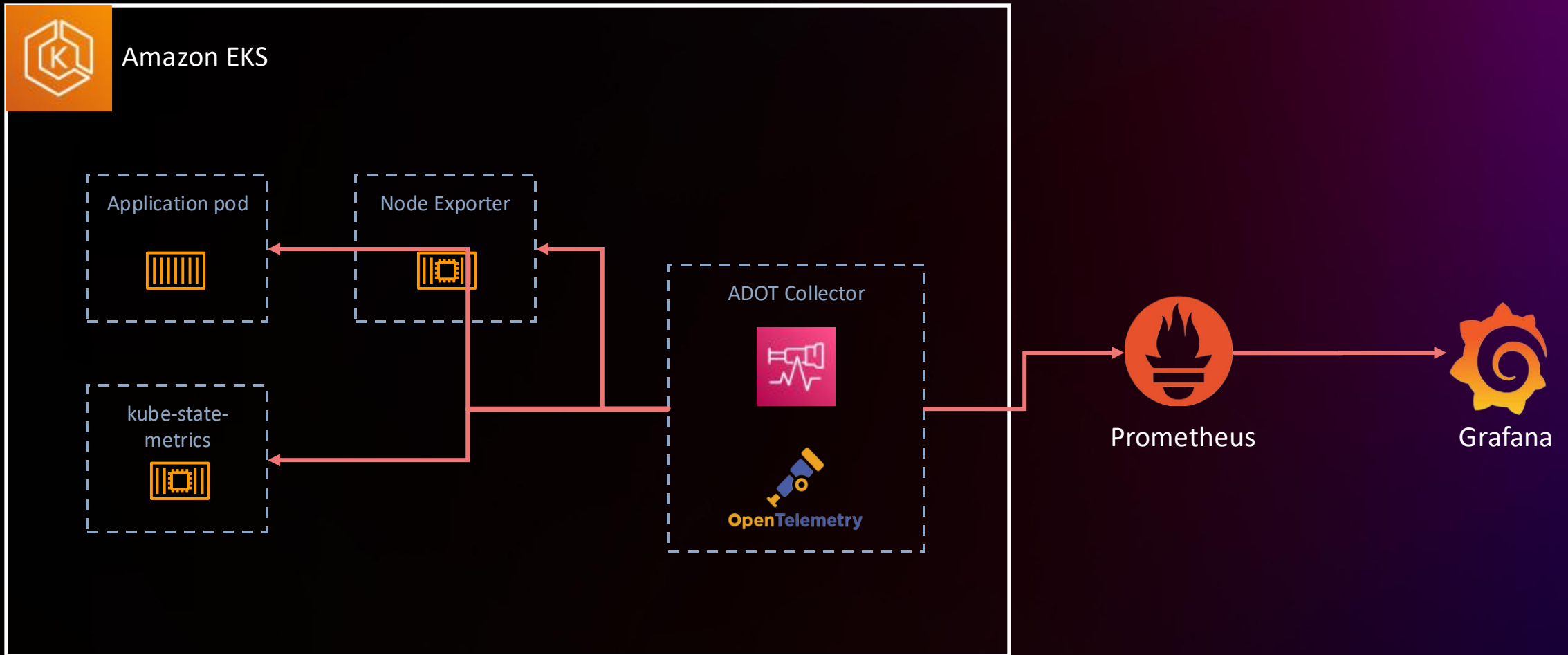


Spans timeline

Group by nodes	Segment status	Response code	Duration	Hosted in	
▼ nginx-proxy					
nginx-proxy	OK	200	4.58s		
▼ pet-clinic-frontend-java					
pet-clinic-frontend-java	OK	200	4.58s	pet-clinic-fronten...	
ApiController.addPaym...	OK	-	4.58s	pet-clinic-fronten...	
ApiController.addPa...	OK	-	4.58s	pet-clinic-fronten...	
PaymentClient.ad...	OK	-	4.58s	pet-clinic-fronten...	
payment-servic...	OK	200	4.58s	pet-clinic-fronten...	Remote: POST http://192.168.36.224:8085/owners/6/pets/7/payments
▼ payment-service-dotnet					
payment-service-dotnet	OK	200	4.56s	payment-service-...	
SQS	OK	200	19ms	payment-service-...	§getQueueUrl
SQS	OK	200	11ms	payment-service-...	§sendMessage: https://sqs.us-east-1.amazonaws.com/140023401067/audit-jobs
customers-service-java	OK	200	4.47s	payment-service-...	Remote: GET http://192.168.45.58:8081/owners/6/pets/7
DynamoDB	OK	200	6ms	payment-service-...	
▼ SQS AWS::SQS					
SQS	OK	200	19ms		§getQueueUrl
▼ SQS AWS::SQS::Queue					
SQS	OK	200	11ms		§sendMessage: https://sqs.us-east-1.amazonaws.com/140023401067/audit-jobs
QueueTime	OK	-	11ms		

- **Configure OTLP collector** : Set up your OpenTelemetry collector to send data to CloudWatch by configuring it to use the correct endpoint and authentication.
- **No language barrier**: Supported from any programming language(s) with OTLP

ADOT with open source observability



Amazon Managed Service for Prometheus



HIGHLY AVAILABLE, SECURE, AND MANAGED METRIC MONITORING

A serverless Prometheus-compatible monitoring service for metrics to securely monitor container environments at scale

Fully managed, secure, and highly available using multi-AZ deployments

Use the same open source Prometheus data model and query language as they do today to monitor the performance of their containerized workloads

No up-front investments required to use the service, and customers only pay for the number of metrics ingested

Improved scalability, availability, and security without having to manage the underlying infrastructure



Amazon Managed Grafana



Scalable, secure and highly available Fully-managed Grafana service.

Automatic scaling, helps offloads operational management

Analyze, monitor, and alarm across multiple data sources; native AWS as well as 3rd party

Native integration with multiple AWS Services for Enterprise-ready security

Access to Grafana Enterprise data source via AWS marketplace directly from the console

Simple pay-as-you-go AWS billing





AI for Observability

AWS DevOps Agent

- "Why is the catalog service slow?"
- "Check the health of my EKS pods"
- "What's causing high CPU on the catalog service?"
- "Show me recent errors in the orders service logs"
- "Why is the website not loading?"
- "Are there any pod restarts or OOMKills?"



Challenges



Increase in application complexity



Telemetry spread across many tools



Fragmented expertise with distributed systems



Firefighting vs systematic improvements

Top Nodes per CPU Utilization

1 minute ▾

Average ▾

1h

3h

12h

1d

3d

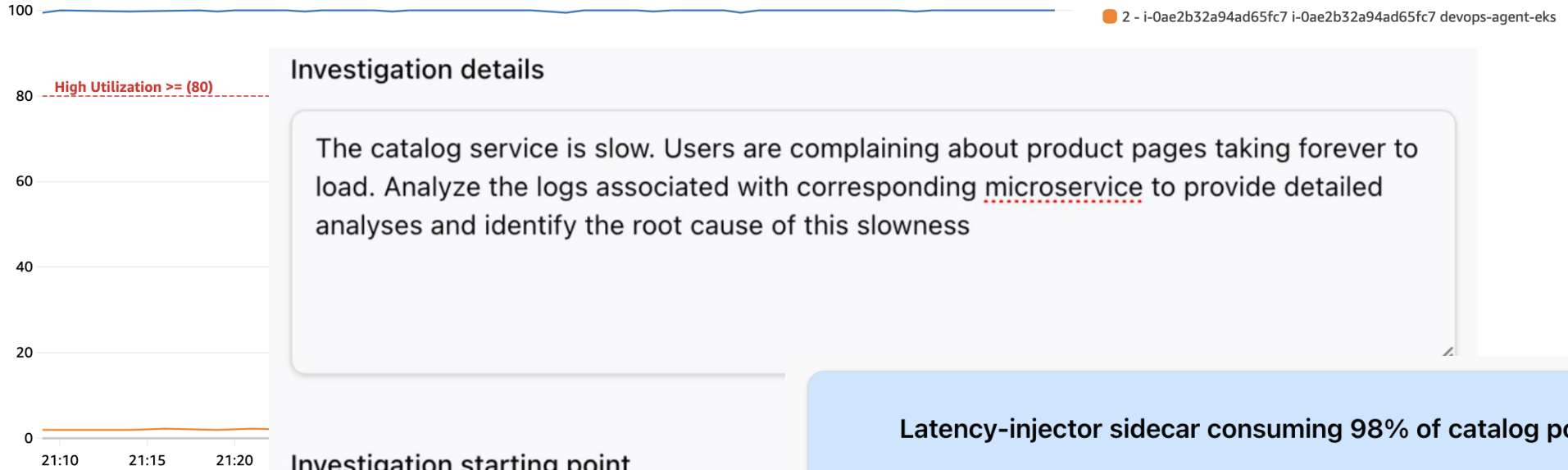
1w

Custom 

UTC timezone ▾



Percent



Investigation details

The catalog service is slow. Users are complaining about product pages taking forever to load. Analyze the logs associated with corresponding microservice to provide detailed analyses and identify the root cause of this slowness

Investigation starting point

Retail website hosted on EKS cluster is complaining about multiple issues incl

Latency-injector sidecar consuming 98% of catalog pod CPU resources

A latency-injector sidecar container is intentionally deployed in the catalog pod (catalog-f5cfbd6fb-z8xct) and is monopolizing CPU resources by consuming 97-98% of the pod's CPU throughout the incident period. The actual catalog application container is receiving only 0.003-0.015% CPU, effectively starving it of compute resources. This CPU exhaustion directly causes the catalog service slowness, as the application cannot process requests efficiently. The sidecar logs confirm it is running 'Latency + CPU stress' intentionally. This appears to be a chaos engineering component that has been active since at least 2026-01-09T20:45:49Z (before the baseline window), indicating this is a pre-existing condition rather than a new incident at 2026-01-09T22:14:16Z.

[Go to root cause](#)



AWS DevOps Agent

Drive operational excellence with a frontier agent that resolves and proactively prevents incidents

Always-on, autonomous incident response

Identify **Root Cause**, Recommend Improvement

Access untapped **insights** in your operations and workloads

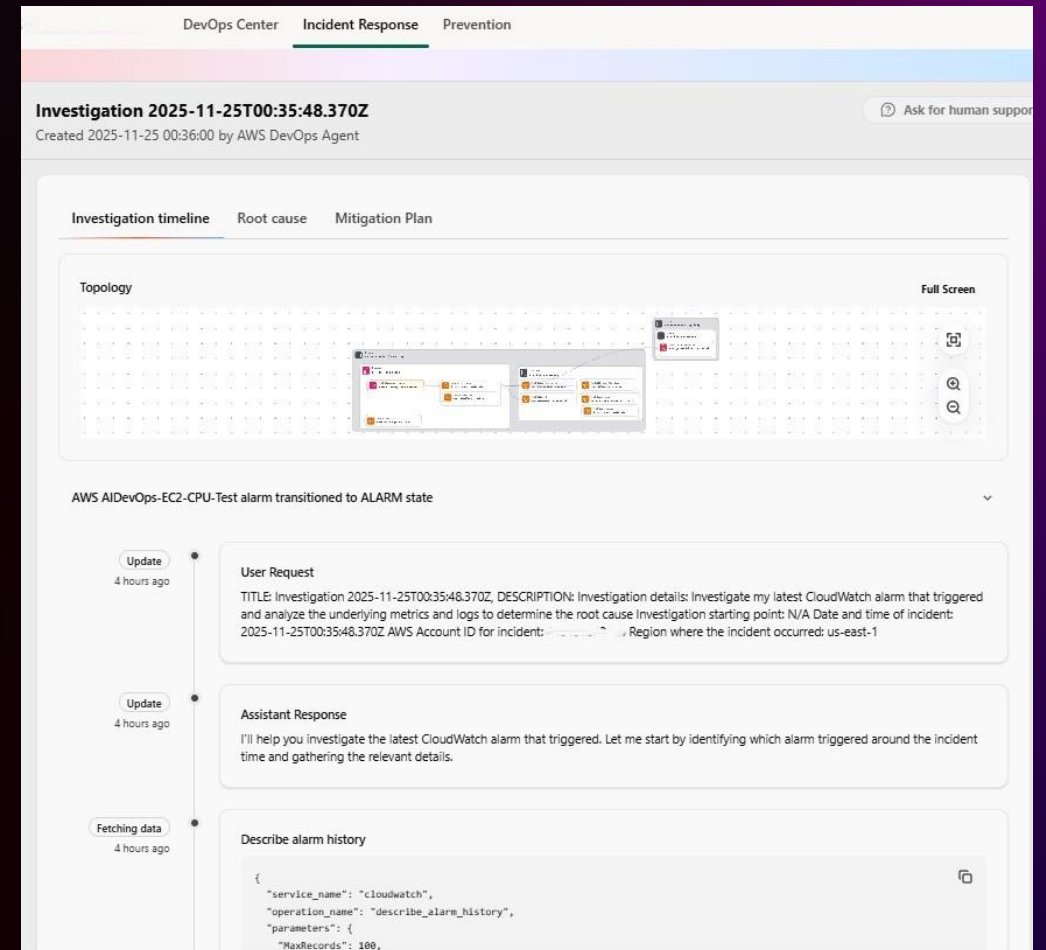
Autonomous incident response

Investigates operational alarms and incident tickets via webhooks

Correlates and analyzes telemetry, code, and deployment data to identify root causes

Works within your collaboration tools like ServiceNow and Slack

Provides detailed mitigation plans



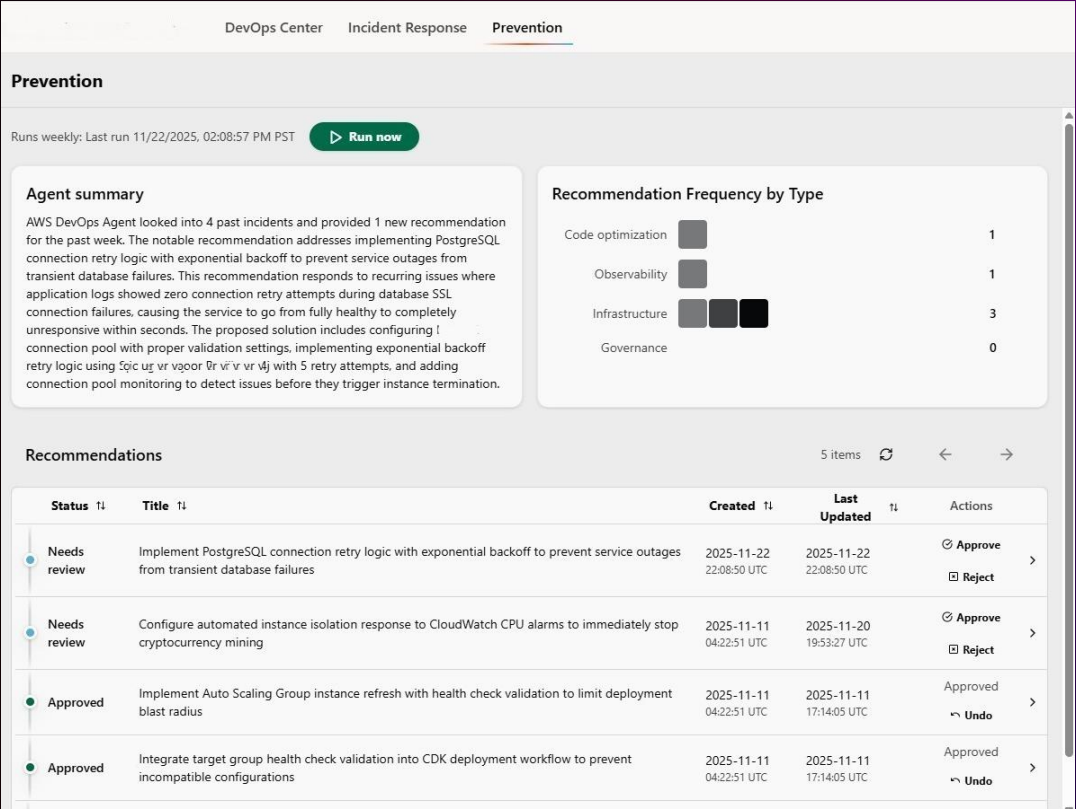
Proactively prevent future incidents

Move from reactive firefighting to proactive operational improvement

Analyzes historical incidents

Identifies gaps in observability coverage to reduce mean time to detect

Provides targeted recommendations to prevent future incidents and strengthen system resilience



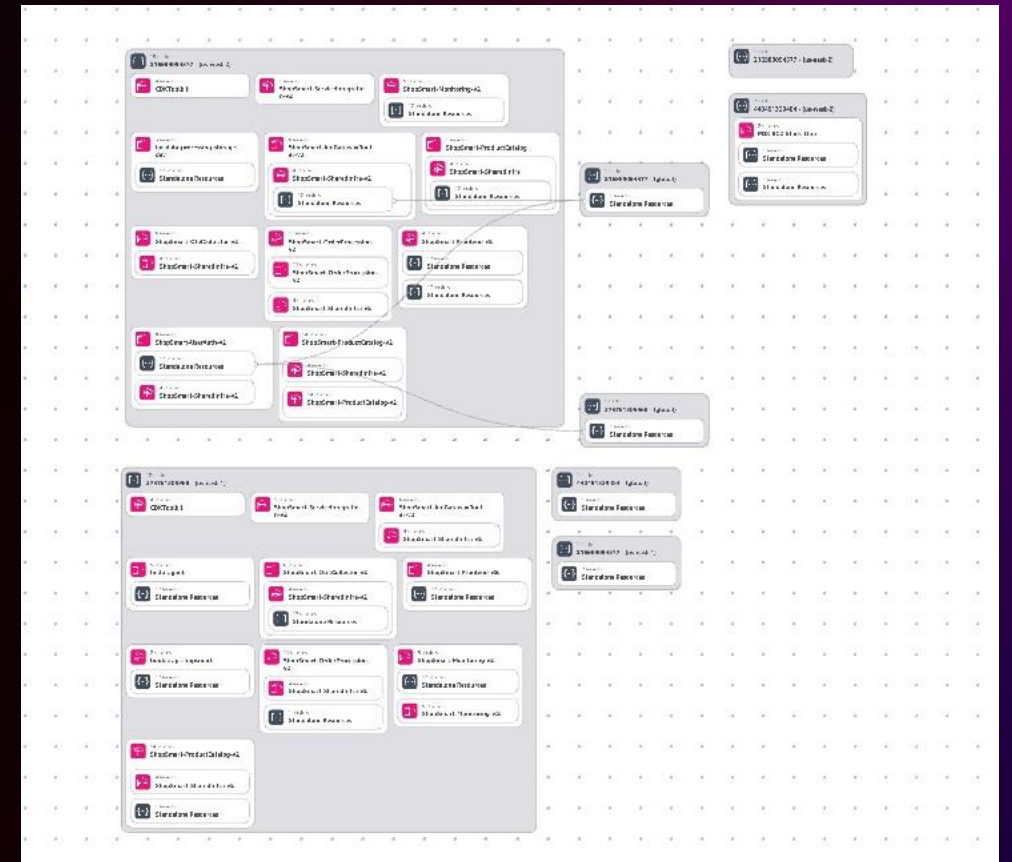
Access untapped insights in your operations and workloads

Understands your application resources and their relationships

Integrates with your observability tools, runbooks, code repositories, and CI/CD pipelines

Built-in integrations with AWS, Dynatrace, DataDog, NewRelic, Splunk, GitHub, and GitLab

Extensible to additional tools by connecting to your own MCP server





AWS DevOps Agent

Autonomous AI Teammate for Observability & Incident Response



Core Capabilities



Automated Incident Response

Autonomous investigation triggered by alerts from CloudWatch, tickets, or monitoring tools



Root Cause Analysis

Correlates telemetry, code changes, and deployment data to identify root causes

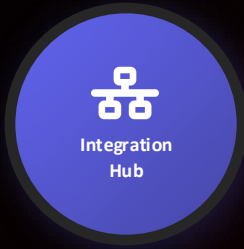


Proactive Reliability

Analyzes patterns to recommend improvements in observability and architecture

Integration Status

Active



DataDog

Cloud Monitoring

- ✓ 90+ AWS service integrations
- ✓ CloudWatch metrics & logs
- ✓ Container monitoring (ECS/EKS)
- ✓ Auto-generated dashboards



Dynatrace

Full-Stack Observability

- ✓ AI-powered root cause analysis
- ✓ Automatic topology mapping
- ✓ Real-time performance monitoring
- ✓ Davis AI for anomaly detection



New Relic

Intelligent Observability

- ✓ MCP Server integration
- ✓ Automated investigations
- ✓ Business context correlation
- ✓ Distributed tracing



Splunk

Data Analytics & Security

- ✓ Splunk MCP integration
- ✓ Real-time data analysis
- ✓ Security & observability unified
- ✓ AI Toolkit with SageMaker

90+

AWS Services

24/7

Always-On AI

MTTR

Reduced Resolution

MCP

Protocol Support



DEMO



Demo - Inject performance issues:

- Reduces main container CPU limits from 256m to 128m (throttling)
- Adds a stress sidecar with CPU workers at 100% load

Simulates

- Runaway processes consuming
- Resource contention issues
- Containers impacting main application

K8s Monitoring & Alert Architecture Flow

